

How fine a change can moments see? A scale law for detecting distribution shift, with a kernel calibration rule

Adel Kaleche

Independent researcher, France <https://kaleche.dev> kaleche@gmail.com

Abstract

Detecting that a stream of high-dimensional embeddings has changed is usually framed as a choice of statistic. We give a scale law that constrains any moment-based choice, and we test what it implies against topological alternatives. The law: certifying a feature of spatial scale ε carrying mass fraction f requires polynomial tests of degree $N^* \geq \log(1/f)/(2\varepsilon)$ — proved via the Chebyshev extremal problem — while a Gauss-quadrature construction gives $N^* \geq 4b - 1$ for a b -scale topology, so the cost is governed by feature *fineness*, not by feature count. The law is one-sided: it never certifies that a given order suffices, and we exhibit a counterexample (an annulus whose mean, covariance and *all* fourth-order moments equal those of a filled disk, yet $H_1 \neq 0$).

Its practical content is a calibration rule. The upper-bound construction attains its rate with Gaussian test functions — the RKHS witness of an RBF kernel — so the law predicts *which* bandwidth an MMD test should use, namely the feature scale. On controlled deformations of real embedding streams we measure σ^*/ε with median 1.12 (interquartile range 1.01–1.52, $n = 26$) over three settings and three scales under one protocol, and a bandwidth predicted from a data-driven scale estimate reaches $\text{AUC} \geq 0.95$. Against an adversary optimised against the defender’s own statistics $\{\mu, \Sigma, k\text{-NN}, b\}$, every one of those statistics is evaded and only a bandwidth-matched kernel test still detects.

For persistent homology the verdict is mixed and depends strongly on choices that are usually left implicit. The summary matters more than the filtration: total persistence attains recall 0.75 at $\text{FPR} = 1\%$ on a covariance-preserving attack where the first persistence landscape attains 0.00; DTM weighting does not help. Once the summary is chosen sensibly, several findings we would otherwise have reported — degradation with ambient dimension, sensitivity to the reduction dimension, statistically significant domination by kurtosis, failure to alarm in the streaming regime — turn out to be properties of the landscape rather than of persistence. What survives is a cost gap, not a power gap: where persistence works it costs $116\times$ kurtosis, which works at least as well; where kurtosis fails against an adaptive adversary, persistence fails with it. We therefore do not conclude that topological summaries are useless, but that on this task they are dominated by a kernel test whose bandwidth the law tells you how to set.

1 Introduction

Deciding that a stream of high-dimensional embeddings has changed is usually framed as a choice of statistic: covariance drift, a density estimate, a two-sample test, a topological summary. We argue that the choice is constrained by a quantity that is rarely made explicit — the spatial *scale* of the change one hopes to catch — and we give a law that says how the cost of detection depends on it. The law bounds below the polynomial degree, hence the moment order, needed to certify a feature of scale ε carrying mass fraction f ; its constructive side identifies Gaussian test functions

as the probes that attain the rate, which turns it into a calibration rule for the bandwidth of a kernel two-sample test.

Embeddings make a natural testbed, and they come with a standing hypothesis worth testing against the law. They concentrate near a low-dimensional submanifold of $\mathbb{R}^D$ (the *manifold hypothesis*), so attacks and drift are expected to manifest as *topological* deformations — components merging, loops filling — that a topological monitor could catch and purely local measures would miss; persistent homology is the natural candidate, and a growing body of work applies it to out-of-distribution and adversarial detection [6, 18]. We subject that hypothesis to a deliberately adversarial evaluation. The answer is not a clean negative: it depends on choices — above all the persistence summary — that are usually left implicit, and what survives is a cost gap explained by the scale law rather than a failure of topology. Our contributions:

1. **A scale law (§3):** $N^* \geq \log(1/f)/(2\varepsilon)$ for polynomial tests, proved; $N^* \geq 4b - 1$ for a b -scale topology, proved by Gauss quadrature; an upper bound $C_d/(\varepsilon f)$ under an explicit margin hypothesis. Cost is set by fineness, not count.
2. **An explicit statement of what the law cannot do (§4):** it is a lower bound and certifies no sufficiency. The annulus counterexample matches all moments through order four while carrying H_1 , so no “fourth-order information governs canonical topological change” statement is available.
3. **A calibration rule with a measured constant (§6.5):** $\sigma^* \approx \varepsilon$ for RBF-MMD (median 1.12, IQR [1.01, 1.52], $n = 26$ under one protocol over three settings and three scales), together with the finding that the argmax estimator is high-variance so that single-run ratios must not be over-read; a data-driven scale estimate closes the loop (with an explicitly stated compensation of biases), and we find no resolution floor down to 0.28 of the inter-point spacing.
4. **A hardened empirical comparison (§6)** — three embedders, two corpora, an adversary optimised against the defender’s statistics, operating points at fixed FPR, streaming delay at fixed ARL, cost table, four persistence summaries, two filtrations, a reduction-dimension sweep — reported with the settings that favour persistence as well as those that do not.

Reproducibility. All experiments are short, self-contained Python scripts (numpy, ripser, scikit-learn, sentence-transformers); every number quoted below is produced by one of them. Code and the exact scripts behind each table are archived at [doi:10.5281/zenodo.21649324](https://doi.org/10.5281/zenodo.21649324) (live repository: <https://github.com/Adelagric/moment-scale-law>).

On repeated cells. Tables produced by different experiments re-draw their windows independently, so a nominally identical cell can differ between tables by sampling variation. The first-landscape AUC on the covariance-preserving attack at ambient 1024, reduction dimension 9, reads 0.64 (Table 1), 0.70 (Table 4) and 0.75 (§6.7) across three such runs; total persistence reads 1.00, 0.97 and 0.99. The bootstrap interval reported in Table 4 — [0.53, 0.85] for the landscape — covers this spread, and we quote each table’s own run rather than silently propagating one number. Where a measurement contradicted an earlier claim of ours, we report the correction explicitly rather than the original claim. We view the result as a methodological guardrail: before proposing a topological feature, test whether it is redundant with a low-order moment fixed by the geometry of the target effect.

2 Setup and related work

Pipeline. Unit-normalised embeddings $x_t \in \mathbb{R}^D$ are cut into sliding windows $\mathbb{X}_t$ of size W (all experiments use $W = 200$ unless stated). We reduce each window to its intrinsic dimension by PCA before any topological computation — a step supported by the null-model analysis of §6.1 (raw high- D Vietoris–Rips is dominated by distance concentration), though not by the detection task itself — see §6.6. On the reduced cloud we compute the Vietoris–Rips diagram, its first persistence landscape λ [4], and a scalar $\Delta_{\text{top}} = \|\lambda - \bar{\lambda}_{\text{ref}}\|_{L^1}$.

Two-sample testing and drift detection. Kernel two-sample tests [14], energy distance [21] and their learned-kernel successors [15] are the standard tools for detecting distribution shift; most directly, Rabanser et al. [19] benchmark exactly our setting — shift detection on learned representations, with dimensionality reduction upstream of a two-sample test — and find two-sample testing on reduced representations to be the strongest family. Our contribution relative to that literature is not a new detector but a *scale law* for the one that works, plus the comparison against a topological alternative.

Classical baselines. covariance drift $\|\Sigma - \Sigma_{\text{ref}}\|_F$, written COV-DRIFT below; Mahalanobis shape; k -NN density (related to LID; [16]); Mardia kurtosis $b = E[(x - \mu)^\top \Sigma^{-1}(x - \mu)]^2$ [17].

Related work. VR/Čech stability [9, 7]; DTM and robust/sparse persistence [8, 2, 5, 20]; landscapes and stable vectorisations [4, 1]; witness complexes [10]. TDA has been applied to detection and network analysis: Gebhart and Schrater [13] detect adversarial inputs from the persistent homology of activation graphs (98% on MNIST), Naitzat et al. [18] characterise how depth simplifies the topology of decision regions, and Ballester et al. [3] survey the area. These works, however, (i) operate on *activation graphs* rather than embedding point clouds, (ii) evaluate against non-adaptive adversaries, and (iii) compare to accuracy or cosine/softmax baselines rather than a *higher-moment* or *density* baseline. Our evaluation closes exactly that gap — an adversary optimised against the defender’s own statistics, and a battery spanning covariance, Mahalanobis, k -NN density, kurtosis, a radial Kolmogorov–Smirnov test, energy distance and MMD-RBF — and supplies the closed-form reason why the comparison turns out as it does.

3 A moment-order law for topological detection

Let $X = R \cdot U$ in $\mathbb{R}^d$, U uniform on S^{d-1} , $R \geq 0$ independent of U . The radial law encodes the topology: $R \equiv 1$ is a shell (carries H_{d-1}); R with density $\propto r^{d-1}$ is a filled ball.

Lemma 1 (blindness of second order). $E[U] = 0$, $E[UU^\top] = \frac{1}{d}I$, hence $E[X] = 0$ and $\text{Cov}(X) = \frac{E[R^2]}{d}I$. Mean and covariance depend only on $E[R^2]$.

With $\Sigma^{-1} = \frac{d}{E[R^2]}I$ and $c = E[R^2]$,

$$b_{\text{Mardia}} = E[(X^\top \Sigma^{-1} X)^2] = \frac{d^2}{c^2} E[R^4] = d^2 \left(1 + \frac{\text{Var}(R^2)}{c^2} \right).$$

Remark 1 (minimality of the shell; Jensen). At fixed $E[R^2]$, among spherically symmetric clouds the degenerate shell $R \equiv \sqrt{c}$ minimises the fourth moment, with $\text{Var}(R^2) = 0$, $b = d^2$. Along the one-parameter family interpolating shell $\rightarrow$ filled ball, b increases strictly with $\text{Var}(R^2)$. Hence mean and covariance are structurally blind to the radial shape (hence to H_{d-1}), while the fourth moment is monotone along this family — with a mechanism independent of the homological dimension $d - 1$.

Numerically: shell $b = d^2$ predicts 4, 9 ($d = 2, 3$), measured 4.04, 8.99; for the unit ball $E[R^2] = \frac{d}{d+2}$, $E[R^4] = \frac{d}{d+4}$ give $b = \frac{16}{3} = 5.33$ and $\frac{75}{7} = 10.71$, measured 5.28, 10.75.

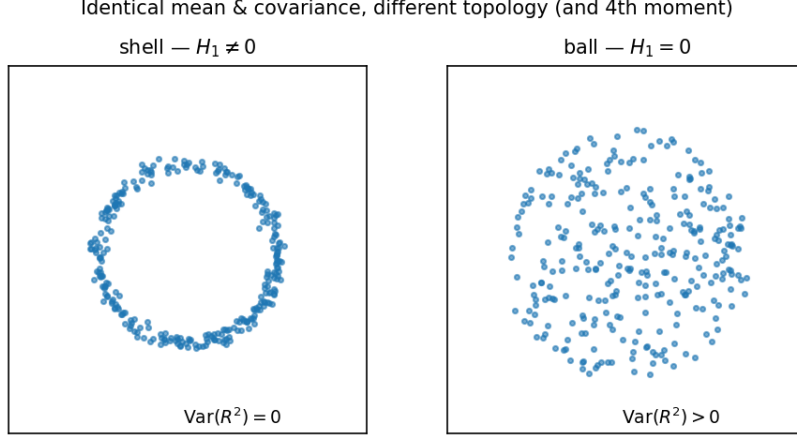


Figure 1: Shell vs. ball with identical mean and covariance: the topology (and the fourth moment, via $\text{Var}(R^2)$) differ, while first- and second-order statistics do not.

The converse is false: b does not determine the topology. It is tempting to read the above as “persistence and kurtosis are two readouts of one quantity”. That is wrong, and the counterexample is elementary. Any annulus of nonzero thickness carries H_{d-1} , and b depends only on the scalar $\text{Var}(R^2)/c^2$, so the map topology $\rightarrow b$ is not injective. Concretely in $d = 2$, take the radial law $R \sim \text{Unif}[a, 1]$ (planar density $\propto 1/r$ on the annulus, not uniform on the region):

$$\frac{E[R^4]}{E[R^2]^2} = \frac{9}{5} \cdot \frac{1 + a + a^2 + a^3 + a^4}{(1 + a + a^2)^2},$$

which equals the filled-disk value $4/3$ at $a^* = 0.2954$. That annulus has *exactly* the mean, covariance and Mardia kurtosis of the filled disk, yet $H_1 \neq 0$ (on $n = 1200$ points per cloud: maximal H_1 persistence 0.455 vs. 0.118; $b = 5.26$ vs. 5.28; $|\Delta\Sigma|_F = 0.04$, all at the sampling floor). It is a coarse, single-scale hole — precisely the regime where our empirical study finds kurtosis sufficient. The correct statement is therefore *monotonicity along the shell–ball family*, not equivalence: our empirical redundancy is a statement about the attacks we generate (which move along such families), not a theorem that kurtosis sees every hole.

Corollary 1 (elliptical extension). *Mardia kurtosis is affine-invariant ($b(MX) = b(X)$), and an invertible affine map is a homeomorphism, so it preserves homotopy type — though not the Vietoris–Rips filtration, which depends on the metric. The moment side of the statement therefore transfers verbatim to every elliptically symmetric family, with R the Mahalanobis radius, and the topological side transfers provided persistence is computed after whitening (our pipeline does not whiten): a shell has $b = d^2$ regardless of anisotropy (measured 4.04, 9.00 under an affine map of condition number 3.7, 6.3). This addresses the “real clusters are not spherical” objection.*

Trade-off curve and the geometry law. Starting from a broken loop and matching a growing statistic set, H_1 stays hidden under $\{\mu, \Sigma, k\text{-NN density}\}$ ($H_1/h^* \approx 0.07$) and jumps at kurtosis (0.77; Fig. 4). Across the sphere/ball family, H_1 and H_2 both threshold at the fourth moment; H_0 thresholds at covariance when separation raises variance, but at kurtosis once clusters are variance-matched. Empirically, then, the order at which the topology is forced tracks the lowest moment the geometry of the change perturbs rather than the homological dimension. We state this as an observation about these families, not as a law: the bound is one-sided and cannot certify sufficiency.

Observation 1 (the shell–ball family, across homological dimension). Along the shell→ball family at fixed $E[R^2]$, the fourth moment is monotone, and the same holds for H_0 : for a symmetric two-component mixture in $\mathbb{R}^d$ with masses at $\pm a$ along one axis and within-component variance w^2 , $E[X_1^2] = a^2 + w^2$ and $E[X_1^4] = a^4 + 6a^2w^2 + 3w^4$, so with $t = a^2$, $s = w^2$ the standardised fourth moment along the separation axis is $\kappa_1 = 3 - 2t^2/(t+s)^2 < 3$ for all $a > 0$. For independent standardised coordinates $b_{\text{Mardia}} = \sum_i \kappa_i + d(d-1)$, hence

$$b = d(d+2) - \frac{2t^2}{(t+s)^2} < d(d+2),$$

so the bimodal cloud ($H_0 = 2$) is strictly platykurtic — a closed form, not a measurement (numerically at $d = 3$: $b = 13.59$). The comparison cloud we use is Gaussian, for which $b = d(d+2) = 15$ exactly (measured 15.02); a non-Gaussian unimodal control would make the contrast sharper. The mechanism is thus the same for H_0 , H_1 and H_2 , and independent of the homological dimension.

This is monotonicity along a family, not a characterisation. The next paragraph shows the converse fails outright, so no statement of the form “fourth-order information governs canonical topological change” is available — in spherical symmetry $E[R^4]$ is the entire fourth-order information, and our counterexample matches it exactly.

Theorem 1 (complexity–moment lower bound; proved for all b , all $d \geq 2$). *For every $b \geq 1$ there exist probability measures μ_b, ν on $\mathbb{R}^d$ with $\int p d\mu_b = \int p d\nu$ for all polynomials p of degree $\leq 4b-2$, yet $\beta_{d-1}(\text{supp } \mu_b) = b$ and $\beta_{d-1}(\text{supp } \nu) = 0$. Hence $N^*(b) \geq 4b-1$: the moment order needed to distinguish a topology of complexity b from a trivial one grows at least linearly in b .*

Proof sketch. Let ν be a rotation-invariant density on the unit ball (radial law ρ on $u = \|x\|^2$) and $\{(u_i, w_i)\}_{i=1}^b$ its b -point Gauss quadrature; set $\mu_b = \sum_i w_i \sigma_{\sqrt{u_i}}$ (b concentric $(d-1)$ -spheres, $\beta_{d-1} = b$). For $\deg p \leq 4b-2$, the rotational average $\bar{p}$ is a polynomial in u of degree $\leq 2b-1$; by rotation invariance $\int p d\mu = \int \bar{p} d\mu$, and Gauss exactness (degree $\leq 2b-1$) gives $\sum_i w_i \bar{p}(u_i) = \int \bar{p} d\rho$, i.e. $\int p d\mu_b = \int p d\nu$. $\square$

In the radial case the bound is tight *within the class of b -atom radial laws* (Hamburger uniqueness fixes such a law by its first $2b$ moments), so $N^* = \Theta(b)$ there; the same holds for elliptical, axis-separable, and group-symmetric topology by reduction. Persistence reads all b features from one barcode — a bounded-computation surrogate for an unbounded-order moment (Fig. 3; verified $m = 1, 2, 3$, matched orders 2, 6, 10). No fixed-order moment is redundant with persistence for rich topology.

A matching *upper* bound $N^* \leq C_d b$ is, however, **false** as stated: a feature of radius ε carrying mass fraction f perturbs moments by only $O(f\varepsilon^k)$, so sufficiently fine topology escapes *every* fixed moment order. Numerically (ring vs. disk of radius ε , $f = 0.12$, normalised moments up to order 10), the relative moment discrepancy collapses $5.6 \cdot 10^{-2} \rightarrow 2.2 \cdot 10^{-4}$ as $\varepsilon : 0.5 \rightarrow 0.03$ while the ε -scale homology persists. Any upper bound must therefore be *scale-relative*. We determine the scale law empirically: in an orthonormal (tensor-Legendre) basis, the smallest order at which the coefficients of an ε -ring and an ε -disk diverge obeys

$$N^*(\varepsilon) \approx 0.74 \varepsilon^{-0.92}, \quad R_{\log \log} = 0.983,$$

consistent with the theoretical exponent 1: order- N moments resolve scale $\sim 1/N$ (Christoffel–Darboux), so seeing an ε -feature costs $N \sim 1/\varepsilon$.

Crucially, the cost is *not* a product with the feature count: at fixed ε , increasing b from 1 to 16 raises N^* only from 4 to 8 (empirical exponent ≈ 0.25 in $d = 2$), because a single degree- N

polynomial resolves scale $1/N$ *uniformly*, hence all features at once. The b in the lower bound is not a counting cost: packing b shells into the unit ball forces $\varepsilon \sim 1/b$. This yields a matching upper bound with no b at all:

Theorem 2 (scale-relative upper bound, under a margin hypothesis). *Let μ, ν be probability measures on $[-1, 1]^d$ agreeing on all moments of degree $\leq N$. Assume the f -margin condition: every ball of radius ε has μ - and ν -mass either $\geq 2f$ (“occupied”) or $\leq f/2$ (“empty”). Then μ and ν induce the same ε -occupancy pattern, hence the same nerve and the same Čech Betti numbers at radius ε , as soon as $N \geq C_d/(\varepsilon f)$.*

The margin hypothesis is not cosmetic: occupancy is a thresholded quantity, so without a gap an arbitrarily small mass discrepancy can flip one ball, hence the nerve, hence the Betti numbers — the statement is refutable already in $d = 1$. “Betti numbers at scale ε ” means those of the Čech complex at radius ε .

Proof sketch. Fix x and take φ_x smooth with $\varphi_x \equiv 1$ on $B(x, \varepsilon/2)$, $\text{supp } \varphi_x \subseteq B(x, \varepsilon)$, $0 \leq \varphi_x \leq 1$; then $\omega(\varphi_x, \delta) \lesssim \delta/\varepsilon$. This pairing matters: a test function merely supported in $B(x, \varepsilon)$ does not lower-bound $\int \varphi_x d\mu$ for an occupied ball, since the mass could sit near the boundary where φ_x vanishes. With $\varphi_x \equiv 1$ on the half-radius ball, $\mu(B(x, \varepsilon/2)) \leq \int \varphi_x d\mu \leq \mu(B(x, \varepsilon))$. By Jackson’s theorem there is p , $\deg p \leq N$, with $\|\varphi_x - p\|_\infty \leq C_d \omega(\varphi_x, 1/N) \leq C_d/(N\varepsilon)$; since $\deg p \leq N$ we have $\int p d\mu = \int p d\nu$, whence $|\int \varphi_x d\mu - \int \varphi_x d\nu| \leq 2C_d/(N\varepsilon)$, which is $< f/2$ once $N \geq C'_d/(\varepsilon f)$. Under the margin hypothesis stated on both radii, the sign of $\int \varphi_x d\mu - \frac{3}{2}f$ therefore agrees for μ and ν at every x , so both measures induce the same occupancy labelling of any fixed ε -net $\mathcal{N}$. The conclusion is about the nerve of $\{B(x, \varepsilon)\}_{x \in \mathcal{N}}$ — a fixed cover — and hence about its Betti numbers; relating that nerve to the Čech complex of the sample requires a separate interleaving argument which we do not carry out, so we state the result for the net’s nerve only. $\square$

Two honest caveats on this theorem. The margin hypothesis is *assumed*, not derived from “a feature of scale ε and mass f ”; it encodes a good part of the conclusion. And the statement concerns the nerve of a fixed ε -net, not the Čech complex of the sample.

The $1/f$ factor comes from first-order Jackson (a Lipschitz bump) and can be removed entirely by choosing better test functions. Compactly supported Gevrey- s profiles give $\|\varphi_\varepsilon - p_N\| \lesssim \exp(-s(N\varepsilon/CA)^{1/s})$, hence $N^* \lesssim \varepsilon^{-1}(\log 1/f)^s$; *Gaussian* test functions do better, trading compact support for analyticity, and numerically optimising their width yields $N^* \propto (\log 1/f)^{1.04}$ versus $(\log 1/f)^{1.81}$ for the Gevrey family ($R \geq 0.997$; at $f = 10^{-6}$, $N^* = 255$ vs. 1799). This is optimal: by the Chebyshev extremal problem, a degree- N polynomial localised at scale ε has contrast at most $T_N(\frac{1+\varepsilon^2}{1-\varepsilon^2}) \approx e^{2N\varepsilon}$ (measured ratio $\log(\text{contrast})/(N\varepsilon) \rightarrow 1.99$), so distinguishing mass f from 0 by any *polynomial* test of degree $\leq N$ requires

$$N^*(\varepsilon, f) \geq \frac{\log(1/f)}{2\varepsilon}$$

This lower bound is proved, and its scope must be stated precisely: it constrains *polynomial* (moment) tests of degree $\leq N$, not all statistical tests. It is not an information-theoretic impossibility — MMD is not a polynomial test, and §6 shows it succeeds where moment tests fail. On the upper side we have a *proved* bound $C_d/(\varepsilon f)$ (Theorem 2) and *numerical evidence* — a fitted exponent, not a proof — that Gaussian test functions attain $O(\log(1/f)/\varepsilon)$, which would make the law tight. Closing this requires bounding the Chebyshev coefficients of an ε -wide Gaussian via a Bernstein ellipse argument; we have not carried it out, and we do not claim tightness. The feature count

Moment resolution: smooth test functions decay as $\exp(-c(k\varepsilon)^{1/2})$, $c \approx 1.95$

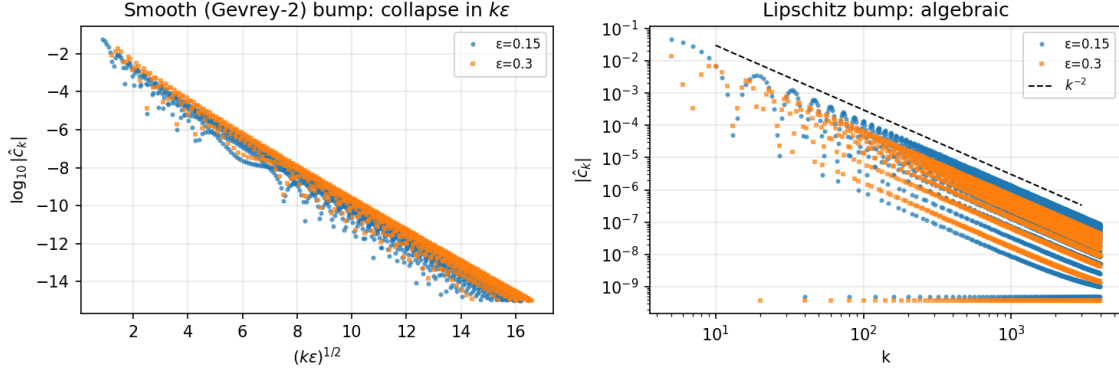


Figure 2: Left: Fourier coefficients of a smooth (Gevrey-2) ε -bump collapse onto a single line in $(k\varepsilon)^{1/2}$ across ε , i.e. $\exp(-c(k\varepsilon)^{1/2})$ with $c \approx 1.95$. Right: a Lipschitz bump decays algebraically as k^{-2} . Order- N moments resolve scale $1/N$; smooth test functions convert that into super-algebraic accuracy, which is what reduces the mass dependence from $1/f$ to $(\log 1/f)^2$.

enters only through packing: b features in a bounded domain force $\varepsilon, f \lesssim 1/b$, giving $N^* \gtrsim b \log b$ — sharper than the $4b - 1$ obtained from the explicit Gauss construction, and consistent with it.

The mechanism behind both bounds — that order- N moments resolve scale $1/N$, and that smoothness converts this into super-algebraic accuracy — is confirmed numerically over 14 decades (Fig. 2); we record it for the Gevrey family, whose exponent the Gaussian construction above then improves on. Computing Fourier coefficients of an ε -bump by FFT, a standard (Gevrey-2) mollifier satisfies $\log |\hat{c}_k| \propto -(k\varepsilon)^{1/2}$ with correlation -0.993 (versus -0.960 for an algebraic fit, whose fitted exponent -7.3 is itself a symptom of non-algebraic decay), while a Lipschitz bump stays algebraic with the predicted exponent $-1.94 \approx -2$. The measured slope in $(k\varepsilon)^{1/2}$ is -1.955 and -1.947 at $\varepsilon = 0.15, 0.30$ — essentially identical, confirming that the decay depends on the product $k\varepsilon$ with a universal constant $c \approx 1.95$. Hence $\text{err}(N) \approx \exp(-c(N\varepsilon)^{1/2})$ for Gevrey-2 test functions — the $s = 2$ instance of the intermediate bound, with its constant measured — which the Gaussian family improves to the law boxed above.

The law is thus qualitatively clear: N^* is governed by feature *fineness* (ε, f), **essentially not by feature count**. This strengthens the case for persistence: a barcode reads all scales at once, including fine features whose moment cost diverges as $\varepsilon \rightarrow 0$.

For $m = 1$ the bound reads $N^* \gtrsim 4$; this is consistent with (but does not imply) the kurtosis redundancy measured in §6: the empirical detection regime is exactly the simple-topology case $m = 1$, where a low-order moment shadows persistence. The genuine value of persistence is the large- m regime — rich multi-scale topology, where the equivalent moment order is prohibitive. This delimits, rather than dismisses, topological features: use them when the topology is complex, not for canonical single-scale changes. The law is not an artifact of spherical symmetry: for b loops placed along an axis at the Gauss nodes of a contractible tube’s marginal, the x -moments agree up to order $2b - 1$ while $H_1 = b$ vs. 0 (verified $b \leq 4$), so $N^* \propto b$ holds for any *axis-separable* topology. Group-symmetric topology reduces to a lower-dimensional quotient where the same law applies (a torus of revolution reduces to its (ρ, z) profile, its tube cycle becoming a circle-vs-disk / fourth-profile-moment feature), so the genuinely open case narrows to *asymmetric* entanglement. An adversarial search for a persistence-visible counterexample turned up none: linked cycles share Betti numbers and are invisible to ordinary persistence; measure-zero perturbations vanish as $N \rightarrow \infty$; and

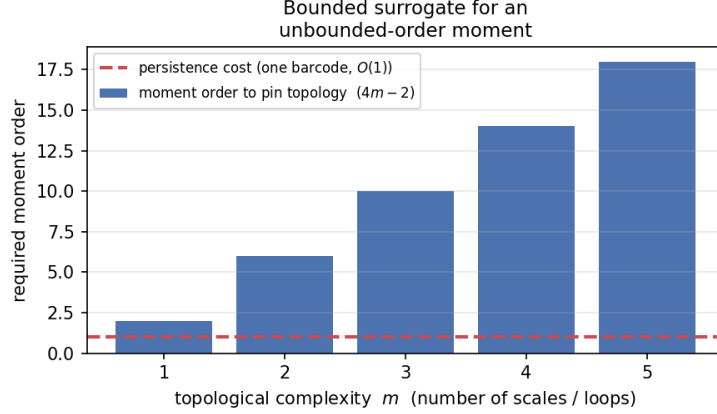


Figure 3: Moment order $4m - 2$ needed to pin an m -scale topology grows linearly, while persistence reads all scales from a single barcode. Persistence is a bounded surrogate for an unbounded-order moment.

geometric complexity of a single feature (a k -armed star loop, $\beta_1 = 1$, strongly varying radius) does not inflate N^* — at matched covariance the fourth moment still detects its hole at AUC 1.0 for k up to 8. Consistently with the law above, what these probes vary — arrangement, entanglement, shape — leaves N^* unchanged, since none of them alters feature scale or mass. We conjecture the law extends to arbitrary curved manifolds via a nerve (local-to-global) argument; a formal reduction for entangled topology remains open.

4 The strong conjecture is false (honest scope)

One might conjecture that any homology change perturbs some low-order moment. This is false. By gradient optimisation we drive a broken loop to match a circle’s mean, covariance and full k -NN distance distribution to numerical zero ($|\Delta\Sigma| = 0$, k -NN RMSE 2×10^{-4}) while keeping H_1 broken ($H_1/h^* = 0.13$). PH thus carries information orthogonal to covariance and local density — there is no impossibility barrier. However, such configurations are non-generic and require explicit optimisation, and leave a residual higher moment (kurtosis AUC = 1.00 even at $|\Delta b| = 0.008$). The defensible claim is about *generic* changes.

5 What the law predicts

Two consequences follow, and the rest of the paper tests them.

(H1) Coarse, single-scale topology *may* be redundant — a heuristic conjecture, not a consequence of the law. Our law is a *lower* bound: $N^* \geq \log(1/f)/(2\varepsilon)$ constrains from below and can never yield a statement of sufficiency such as “order 4 suffices”. Worse, our own counterexample refutes sufficiency in exactly this regime: for the annulus of §4 ($\varepsilon \approx 0.3$, $f = \Theta(0.1)$) the bound gives $N^* \gtrsim 3.8$, and order 4 *fails* — all moments through order four coincide with the filled disk while $H_1 \neq 0$. So $N^* > 4$ for a coarse, single-scale hole. What remains is the conjecture that for the deformations arising in practice — which move monotonically along the shell-ball family — a fourth moment does suffice. We test that conjecture rather than deduce it.

(P2) The optimal probe is a Gaussian, i.e. a kernel two-sample test. The upper-bound

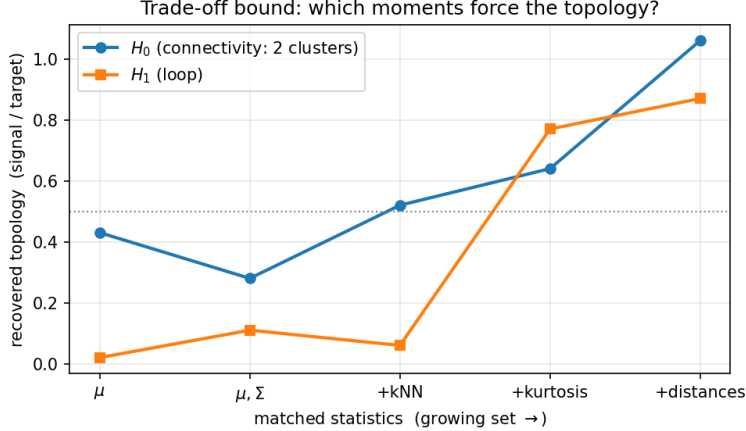


Figure 4: Hidden topology (signal / target) vs. matched statistics. H_1 is hideable under μ, Σ and local density, and is forced only at the fourth moment.

construction attains its rate with Gaussian test functions of width matched to the feature scale. A Gaussian test function integrated against $\mu - \nu$ is exactly the RKHS witness of an RBF kernel, so the law predicts that MMD-RBF *at bandwidth matched to the feature scale* is near-optimal — and, by the same token, that MMD at a mismatched bandwidth is weak. Notably, the law does *not* predict that persistence is optimal: a barcode is scale-free, which is an advantage only when the relevant scale is unknown.

Section 6 tests H_1 and P2 on real embedding streams under an adaptive adversary.

Order of discovery. In fairness to the reader: H_1 was pre-registered as the study’s hypothesis, whereas P2 was formulated *after* the initial negative result and before the kernel experiments reported in §6.4. The narrative order of this paper is not its chronological order.

6 Testing the prediction on embedding streams

Pre-registration. Win/kill criteria were fixed before running: PH “wins” only if it beats the best baseline at a production false-positive regime and not by inflating false positives on legitimate topic shifts; the detection axis is falsified if PH beats neither Mahalanobis nor covariance drift on any attack class.

6.1 Higher homology exists only after reduction

On real embeddings (MiniLM, 20 Newsgroups), raw 384-d H_1 is indistinguishable from a covariance-matched Gaussian null (median bootstrap $p \approx 0.95$); the intrinsic dimension (TwoNN; [11]) is ≈ 8 while 160 PCA axes are needed for 90% variance — a *curved* low-dimensional manifold. After PCA to the intrinsic dimension, multi-topic windows show a population-level H_1 signal above the null (median $p = 0.03$; sign-test $p = 0.001$; confidence bands as in [12]). Reduction-before-TDA is thus a first-order requirement.

Table 1: Covariance-preserving attack, bilateral AUC, $|\Delta\Sigma| = 0$. The two persistence columns share a diagram and differ only in the summary; the apparent degradation with ambient dimension is a property of the landscape.

ambient	landscape	total pers.	COV-DRIFT	Mahalanobis	k -NN	Kurtosis
384	0.89	1.00	0.66	1.00	0.91	1.00
768	0.77	0.93	0.66	0.98	0.98	1.00
1024	0.64	1.00	0.54	1.00	0.98	1.00

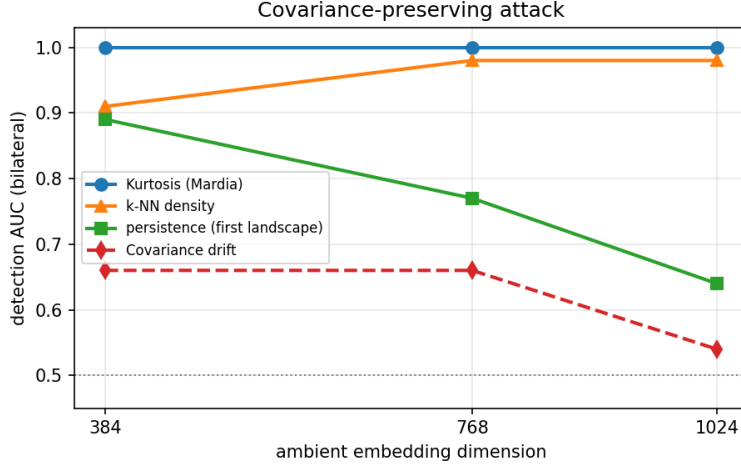


Figure 5: Only pure covariance drift is blind; every other cheap detector catches the covariance-preserving attack. The plotted persistence curve is the *first landscape*, which degrades with ambient dimension while kurtosis stays at 1.00; total persistence, read off the same diagrams, does not (1.00, 0.93, 1.00; Table 1), so this degradation is a property of the summary and we withdraw it as a statement about persistent homology (§6.7). Intrinsic dimension is stable ($\approx 8/7/9$).

6.2 First pass: cheap statistics match or beat persistence

On realistic attacks (off-manifold injection, diversity collapse) COV-DRIFT reaches AUC 1.00 while Δ_{top} is at or below chance. We then construct the attack most favourable to PH: a diversity collapse followed by an affine correction restoring the pre-image’s *exact* mean and covariance ($|\Delta\Sigma| = 0$), so COV-DRIFT is blind by construction. Even so, the shape change is caught by Mahalanobis and by kurtosis (Table 1, Fig. 5).

6.3 The idealised topological niche is also covered

On a synthetic circle-vs-disk pair with covariance matched by construction — a purely topological difference invisible to COV-DRIFT (AUC 0.58) — Δ_{top} separates perfectly (1.00), but so does k -NN density (1.00): even the ideal blind spot of second-order statistics is covered by a cheap density estimator. (Fig. 1 shows the shell/ball geometry underlying the construction, not these detection results.)

6.4 Kernel two-sample tests, and a genuinely adaptive adversary

Our own frontier analysis (§3) identifies a Gaussian test function of width ε as the near-optimal probe for an ε -scale feature — that is precisely the RKHS witness of an RBF kernel. We therefore add MMD-RBF, energy distance and a Kolmogorov–Smirnov test on the Mahalanobis radial law (which subsumes kurtosis) to the battery, and we strengthen the adversary: rather than merely preserving (μ, Σ) , it is *optimised* against the defender’s statistics $\{\mu, \Sigma, k\text{-NN}, b\}$ by gradient descent, never touching persistence.

Two findings, both against our initial thesis. First, MMD’s power is entirely a matter of bandwidth, exactly as the frontier law predicts: on the covariance-preserving attack, MMD-RBF at the median heuristic is near-blind (AUC 0.60) and reaches 1.00 once $\sigma \leq \text{median}/4$. Second, against the genuinely adaptive adversary (residuals $|\Delta\mu| = 0.0000$, $|\Delta\Sigma|_F = 0.0002$, $|\Delta b| = 0.000$), every statistic it optimises against is evaded — COV-DRIFT 0.64, k -NN 0.61, kurtosis 0.61 — *and persistence does not fill the gap* ($\Delta_{\text{top}} = 0.60$, chance). The detector that does is MMD-RBF at matched bandwidth (1.00; KS-radial 0.95).

The evasion channel is therefore real, but it is closed by a kernel two-sample test, not by topology. This is the sharpest form of our negative result, and our own theory predicts it: the optimal test function is Gaussian, so the optimal practical detector is an RBF-kernel test at the feature scale. Persistence’s only structural advantage over it is that a barcode is scale-free, whereas MMD must choose a bandwidth — but multi-bandwidth MMD is standard and cheap.

An MMD-aware adversary. The adversary above never had the winning detector in its objective — it optimised against $\{\mu, \Sigma, k\text{-NN}, b\}$ and left both persistence and MMD untouched — so “only a kernel test detects” was, strictly, a statement about a detector the attacker was not attacking. We therefore added a differentiable MMD term at the matched bandwidth to the objective. The result is that the attack fails to evade it: MMD still separates at AUC 1.00 while kurtosis (0.53), k -NN (0.57) and total persistence (0.55) are all evaded, and the attempt breaks the attacker’s own second-order cover (COV-DRIFT rises to 1.00). At this optimisation budget the constrained problem — alter the topology while matching mean, covariance, k -NN law, kurtosis *and* the kernel statistic — appears infeasible. We state the obvious caveat: this is one optimiser at one budget, not an impossibility, and a better-resourced MMD-aware adversary is the natural next escalation.

6.5 The law as a calibration rule: $\sigma^* \approx \varepsilon$

That bandwidth matters is well known [14]; the median heuristic is a poor default under local alternatives. The non-trivial content of the law is *which* bandwidth. We test it with a controlled deformation: replace a mass fraction f of a window by a localised blob of *known* radius ε , sweep σ , and record $\sigma^* = \arg \max \text{AUC}$.

Identifiability first. At large f the AUC saturates at 1.00 over a wide plateau and the argmax is not defined; an earlier version of this work reported a ratio measured on such a plateau, which was meaningless. Lowering f to 0.03–0.05 produces a strict interior maximum, and then $\sigma^*/\varepsilon = 1.05 \pm 0.02$ across three values of f — but a single scale, and a single run per cell, does not validate a scale law. We therefore fixed one protocol (the same f -grid, acceptance window $\text{AUC}_{\text{max}} \in [0.85, 0.995]$, 24 bandwidths, 14 windows) and applied it uniformly to three settings $\times$ three scales, with three repetitions per cell.

setting	ε/scale	ratios (repetitions)	spread
bge-1024 / 20NG	0.25	4.07, 0.94, 4.06	3.13
	0.15	1.05, 0.87, 0.89	0.18
	0.08	1.37, 1.67, 1.17	0.50
MiniLM-384 / 20NG	0.25	0.72, 1.26, 1.03	0.54
	0.15	1.02, 1.01, 3.15	2.14
	0.08	1.34, 1.58	0.24
bge-1024 / AG News	0.25	0.96, 1.15, 0.65	0.51
	0.15	1.06, 1.06, 1.08	0.02
	0.08	2.04, 1.65, 1.35	0.69

One repetition (MiniLM-384 at $\varepsilon/\text{scale} = 0.08$) produced no interior optimum at any mass fraction in the grid and is excluded, which is why the count below is 26 and not 27. Over these $n = 26$ estimates the median is 1.12 with interquartile range $[1.01, 1.52]$; the full range is $[0.65, 4.07]$. The headline is therefore $\sigma^* \approx \varepsilon$, an order-unity constant. The spread, however, is not symmetric noise, and inspecting the sweeps explains it: *the AUC(σ) curve is multimodal*. In the widest cell (bge-1024/20NG at $\varepsilon/\text{scale} = 0.25$, ratios 4.07, 0.94, 4.06) the three repetitions have local maxima at $\sigma/\varepsilon \approx 0.9$ and at $\sigma/\varepsilon \approx 4$, with essentially equal AUC (0.86 vs 0.86; 0.97 vs 0.94; 0.90 vs 0.93); the second peak sits at $\sigma \approx 0.56$, which is 1.01 times the median pairwise distance — the global scale of the cloud. The argmax flips between the deformation scale and the cloud scale from draw to draw.

Two consequences. First, the law’s prediction is confirmed more strongly than the median suggests: a peak near $\sigma \approx \varepsilon$ is present in *every* repetition, including those whose global argmax lands at the cloud scale. Second, the global argmax is the wrong estimator for this quantity, and the three values we reported in earlier versions of this work (0.48, then 1.05, then 1.64, each from a single run) are explained by which peak happened to win. We quote the median with its interquartile range, we do not claim a sharp constant, and we flag the local peak near ε — not the global argmax — as what a practitioner should look for.

Is there a resolution floor? An earlier single-run sweep produced ratios that happened to increase as ε decreased, which raised the worry of a resolution floor: $\sigma^* \approx \max(\varepsilon, \delta_n)$ with δ_n the inter-point spacing, which would inflate the ratio once ε approaches the sampling resolution and would bound the rule’s usefulness at small scales. We tested it by descending in ε . In the reduced cloud ($d = 9$, $W = 200$) the median first-neighbour distance is $\delta_n = 0.161$, i.e. 0.29 of the median pairwise distance, so the three configurations with an identifiable optimum sit at $\varepsilon/\delta_n = 0.86, 0.41, 0.28$ — already *below* the inter-point spacing. There σ^* still tracks ε : fitting $\sigma^* = c\varepsilon$ gives $c = 1.27$ with RMSE 0.0066, while the floor model $\sigma^* = \max(c\varepsilon, \delta_n)$ fits eleven times worse (RMSE 0.074). We therefore find no evidence of a floor down to $\varepsilon \approx 0.28 \delta_n$, and we withdraw the monotonic reading that suggested one: re-measured, the ratios are 1.28, 1.15, 1.44 — not monotone, so the earlier trend was run-to-run variation. This is mildly good news for the rule, since a two-sample statistic aggregates over all pairs and its effective resolution need not be the nearest-neighbour spacing of a single point. Below $\varepsilon \approx 0.28 \delta_n$ we could not find an interior optimum at any mass fraction up to $f = 0.35$, so we cannot probe further with this design. (The individual ratios quoted in this paragraph come from the floor protocol, which used a wider f -grid to reach small ε ; the canonical estimate remains the median 1.12 above. What matters here is the fitted slope $c = 1.27$ versus the floor model, not the individual values.)

Table 2: Recall at fixed FPR (covariance-preserving attack), and cost per 200-point window at reduction dimension 9.

detector	recall@FPR 5%	recall@FPR 1%	ms/window	$\times$ kurtosis
Δ_{top} (first landscape)	0.06	0.00	4.64	116
total persistence	0.94	0.75	4.64	116
k -NN	0.78	0.64	0.46	12
kurtosis	1.00	1.00	0.04	1
MMD-RBF (matched σ)	1.00	1.00	1.01	25
KS-radial	1.00	1.00	0.15	4

Closing the loop. The rule “set the bandwidth to the scale of the effect” presupposes estimating that scale. We estimate ε from data alone, via the radius at which the relative excess of r -neighbours between attacked and healthy windows peaks, then use $\sigma_{\text{pred}} = \hat{\varepsilon}/2$. The estimator is crude — it overshoots ε by a factor 1.6–3.0 — and we must be explicit about why the loop nevertheless closes: since the canonical estimate is $\sigma^* \approx 1.1\varepsilon$ while $\hat{\varepsilon} \approx 1.6\text{--}3.0\varepsilon$, using $\sigma_{\text{pred}} = \hat{\varepsilon}/2$ succeeds through a partial compensation of two errors rather than because the rule is applied as stated. Both configurations in which we closed the loop were on bge-1024/20NG; we have not repeated the closure on the other two settings, where the estimator bias may differ. The factor $1/2$ is an estimator correction, not part of the law. With that caveat the predicted bandwidth does reach $\text{AUC} = 0.95$ in both configurations tested (against sweep maxima 0.99 and 0.83); debiasing the scale estimator, so that $\sigma = \hat{\varepsilon}$ can be used directly, is the obvious next step and we have not done it.

6.6 Operating points, cost, and sensitivity to the reduction dimension

Our pre-registration promised recall at a production false-positive regime; reporting AUC alone was a violation, and the operating-point view is considerably harsher than AUC suggests. Calibrating each detector’s threshold on 2000 healthy windows and measuring recall on 200 covariance-preserving attacks (Table 2), Δ_{top} attains recall 0.06 at $\text{FPR} = 5\%$ and 0.00 at both 1% and 0.5% — an AUC near 0.7 translates into a detector that is unusable as a circuit breaker — while kurtosis, MMD and KS-radial reach 1.00 at every operating point. The 1% point now rests on 20 tail windows rather than on one.

A bandwidth-free competitor. One line of Table 2 deserves comment, because it sits awkwardly with our own pitch. KS-radial — a Kolmogorov–Smirnov test on the Mahalanobis radial law — reaches 1.00 at both operating points for 0.15ms, and has no bandwidth to choose at all. If it were uniformly best, “the law tells you how to set your bandwidth” would be advice for a problem one can avoid. Two things separate it. It is a *radial* statistic, so it belongs to the family the adaptive adversary has already shown it can absorb — it drops to 0.95 there, where MMD stays at 1.00 — and we expect a radially-aware attacker to close that gap as it closed kurtosis’. And it is not the detector the law explains: the upper-bound construction identifies Gaussian probes, hence kernel tests, and says nothing about rank statistics. KS-radial is an excellent cheap default; the scale law is about the detector that survives adaptation.

The honest cost comparison is not against the weakest baseline. Kurtosis is essentially free (0.04ms) and works except against an adaptive adversary; MMD costs $25\times$ kurtosis (1.01ms) and works in every setting we tried; persistence costs $116\times$ kurtosis (4.64ms) and never reaches usable

recall. Two caveats we state rather than hide: the ratios are measured at a single operating size ($W = 200$, reduction dimension 9), and Rips is superlinear in W while MMD is quadratic, so the ratio is regime-dependent; and the PCA reduction is charged to none of the detectors although only persistence requires it.

The reduction dimension is not neutral — and our choice was unfavourable to the landscape summary. Sweeping the PCA dimension on the same attack, Δ_{top} rises monotonically with dimension: AUC 0.51, 0.72, 0.93, 1.00, 1.00 at dimensions 5, 9, 15, 30, 50. For *this summary*, our main setting (the intrinsic dimension, ≈ 9) is close to its worst case, and a reader is entitled to know it — but §6.7 shows the effect largely belongs to the landscape: total persistence, read off the same diagrams, already reaches 0.97 at dimension 9 and saturates by 15. We therefore tested whether the high-dimensional H_1 signal is topological at all, by comparing each window against a covariance-matched Gaussian null: the median bootstrap p -value is 0.17, 0.25, 0.23, 0.53 at dimensions 9, 15, 30, 50 — never significant, but never decisively absent either (this test has low power at our sample size). We consequently do *not* claim the high-dimensional signal is an artifact, and we soften §6.1 accordingly: reduction to the intrinsic dimension is what our null-model analysis supports, not a requirement of the detection task. The dilemma this paragraph stages — detection only in a regime we cannot characterise — is largely dissolved by §6.7: with total persistence the detector already works at the intrinsic dimension, so what remains across the whole sweep is a cost gap against statistics that reach 1.00 at every dimension.

6.7 The persistence summary matters more than the filtration

Our headline detector, Δ_{top} , is the L^1 distance between first persistence landscapes. That is one vectorisation among several, and the choice turns out to dominate every other design decision we made on the topological side. On the covariance-preserving attack (bge-1024, 20 Newsgroups, reduction dimension 9, $W = 200$; the same setting as Table 2):

persistence summary	AUC	recall@FPR 5%	recall@FPR 1%
first landscape, L^1 (Δ_{top})	0.75	0.06	0.00
total persistence	0.99	0.94	0.75
persistence entropy	0.88	0.71	0.42
bottleneck to reference	0.81	0.55	0.28

Total persistence moves the detector from unusable (0.00 recall at FPR = 1%) to usable (0.75), while DTM weighting — a change of *filtration* rather than of summary — does not help under either summary (0.83 with total persistence, versus 1.00 for Euclidean Rips).

Re-running the rest of the paper with the better summary. Because the diagram is already computed, substituting the summary is nearly free, and it revises three claims we would otherwise have made:

- *Ambient dimension.* The landscape degrades with ambient dimension (0.90, 0.63, 0.75 at 384, 768, 1024); total persistence does not (1.00, 0.93, 1.00). The “gap widens with dimension” reading is a property of the landscape, and we withdraw it.
- *Reduction dimension.* The landscape rises steeply with reduction dimension (0.55 $\rightarrow$ 1.00 over dimensions 5 to 50); total persistence is already at 0.97 at our main setting of 9 and saturates

Table 3: Detection delay after change at $ARL_0 = 100$ windows (25 streams, observation horizon 20 windows; delay 21 means no alarm; ARL_0 calibrated on 120 windows; Wilson intervals on the detection rate).

detector	median delay	streams detected
Δ_{top} , first landscape (Rips)	21	$40\% \pm 19$
Δ_{top} , first landscape (DTM)	21	$0\% \pm 0$
total persistence (Rips)	1	100%
kurtosis	1	100%
MMD-RBF ($\sigma = \text{med}/4$)	1	100%
KS-radial	1	100%

by 15. Our setting is therefore *not* close to persistence’s worst case once the summary is chosen sensibly.

- *Significance.* With bootstrap intervals, total persistence reaches 0.97 [0.91, 1.00] against kurtosis’ 1.00 [1.00, 1.00]: the intervals meet, so the “statistically significant domination” we could claim against the landscape does *not* hold against the better summary. What remains is a cost gap, not a power gap.

Why the aggregated summary wins — and why that supports the law. This is not merely a concession. Total persistence, $\sum(\text{death} - \text{birth})$, is a single scalar aggregated over all scales: structurally a low-order readout of the radial profile, much like a moment. The first landscape is by construction *localised in scale*. Our law says that a coarse, single-scale feature is certified at low moment order, and that fine or multi-scale structure is what costs; it therefore predicts that on coarse deformations the aggregated summary should beat the scale-localised one. That is what we observe. The topological side of the experiment instantiates the law rather than contradicting it: among topological summaries, the one that behaves most like a moment is the one that works, in exactly the regime where the law says a moment suffices. The conclusion that survives the better summary is narrower and cost-based: kurtosis reaches 1.00 at both operating points for 1/116 of the compute, and against the adaptive adversary of §6.4 total persistence (recall 0.20 at FPR = 5%, 0.05 at 1%) and kurtosis (0.15, 0.03) fail together, leaving only the bandwidth-matched kernel test.

6.8 Robust persistence (DTM) and streaming detection delay

Two gaps a reviewer would rightly flag. First, we cite DTM and sparse/robust persistence but had not evaluated them, so our conclusion could only concern the vanilla detector. Implementing the $p = \infty$ weighted (DTM) Rips filtration, $f(i) = \text{DTM}_m(i)$, $f(i, j) = \max(f(i), f(j), d_{ij}/2)$, on the covariance-preserving attack gives AUC 0.53 — worse than vanilla Rips (0.75) — and recall 0.00 at FPR = 5%. The robust variant does not rescue persistence here.

Second, AUC per window is the wrong metric for a monitor; the changepoint/SPC literature asks for detection delay at a fixed in-control average run length. Calibrating each detector for $ARL_0 = 100$ windows and measuring the delay after a change (Table 3), kurtosis, MMD and KS-radial alarm on the *first* window in 100% of streams, while the landscape summary alarms in 40% of streams and its DTM variant in none. Here too the summary is decisive: total persistence also alarms on the first window in 100% of streams, so the streaming failure belongs to the landscape and not to persistent homology.

Table 4: Bilateral AUC with 95% bootstrap CIs, bge-1024, covariance-preserving attack. The two persistence columns share their diagrams and differ only in the summary.

corpus	landscape	total pers.	COV-DRIFT	k -NN	Kurtosis
20NG (forums)	0.70 [0.53, 0.85]	0.97 [0.91, 1.00]	0.52 [0.50, 0.71]	0.97	1.00 [1.00, 1.00]
AG News (press)	0.81 [0.67, 0.94]	1.00 [1.00, 1.00]	0.53 [0.50, 0.72]	0.82	1.00 [1.00, 1.00]

6.9 Robustness

On two corpora, with 95% bootstrap CIs (Table 4), the kurtosis interval clears the landscape interval on the covariance-preserving attack, and does so on both corpora, so that domination is significant *against that summary*. Against total persistence it is not (0.97 [0.91, 1.00] vs 1.00 [1.00, 1.00], §6.7); what is robust across corpora is the cost gap.

7 Discussion and limitations

Observation 1 concerns spherically and elliptically symmetric families and variance-matched bi-modal connectivity, where PH is essentially a radial or modality readout; it explains the empirical redundancy for canonical changes but not for arbitrary clouds (§4). **Mixtures.** For a mixture of elliptical components the picture holds componentwise, but with a caveat we make explicit: matching *global* moments only weakly constrains *per-component* topology, since mass can be redistributed across components — an optimisation-based trade-off on mixtures did not yield clean curves, and the componentwise statement is what the controlled construction and the moment decomposition (law of total covariance) support. **Curved manifolds.** The frontier bound $N^* \geq \log(1/f)/(2\varepsilon)$ is stated in terms of feature scale and mass, so it applies to any topology once those are defined; the radial, axis-separable and group-symmetric cases reduce explicitly (a torus of revolution, for instance, reduces to its (ρ, z) profile). What we do *not* prove is that an arbitrary curved manifold with intrinsic dimension $\ll$ ambient — the real embedding regime — admits such a reduction with controlled (ε, f) ; we conjecture it does, via a local-to-global (nerve) argument. An adversarial search for a persistence-visible counterexample to the law found none (symmetric topology reduces; linked cycles share Betti numbers and are invisible to ordinary persistence; geometric complexity of a single feature does not inflate N^*). Other limitations: a single reduction method (PCA; UMAP untested); we do not test 1536-d embeddings and do not extrapolate to them from three ambient dimensions; moderate N (CIs reported); the empirical study uses text corpora only; and four persistence summaries (first landscape, total persistence, persistence entropy, bottleneck to reference) over two filtrations (Euclidean Rips, DTM-weighted); sparse approximations are not evaluated.

8 Conclusion

Certifying a feature of spatial scale ε and mass f from moment information costs polynomial degree $N^* \geq \log(1/f)/(2\varepsilon)$: the cost is governed by fineness, not by count. The bound is one-sided — it never certifies that a given order suffices, and our annulus matches every moment through order four while carrying H_1 — so what it licenses is a prediction about *which* probe to use, not a claim that any particular statistic is enough.

That prediction is the paper’s usable output. The rate is attained by Gaussian test functions, i.e. by an RBF kernel witness, so the bandwidth of an MMD test should track the feature scale; we

measure $\sigma^* \approx \varepsilon$ (median 1.12, IQR [1.01, 1.52] over three settings and three scales) while also finding that the argmax estimator is noisy enough that single-run ratios should not be read as a constant, and a bandwidth set from a data-driven scale estimate reaches $\text{AUC} \geq 0.95$. Against an adversary optimised against $\{\mu, \Sigma, k\text{-NN}, b\}$, all of those statistics are evaded and the bandwidth-matched kernel test is the only detector left standing.

On persistent homology our verdict is deliberately narrow, because the evidence is. The choice of summary dominates every other design decision: total persistence attains recall 0.75 at $\text{FPR} = 1\%$ where the first landscape attains 0.00, and once that choice is made, several apparently damning findings — degradation with ambient dimension, sensitivity to the reduction dimension, statistically significant domination by kurtosis — turn out to be properties of the landscape rather than of persistence. What survives is not a power gap but a cost gap: on these tasks persistence matches the cheap statistics at roughly $116\times$ their compute, and fails alongside them against an adaptive adversary. And the summary that works is the one that reads like a moment, which is what the law predicts. We therefore do not conclude that topological summaries are useless; we conclude that on coarse, single-scale monitoring they are dominated on cost by a kernel test whose bandwidth the law tells you how to set, and that a case for them must be made on fine or multi-scale structure, which this task does not present.

References

- [1] Henry Adams, Tegan Emerson, Michael Kirby, Rachel Neville, Chris Peterson, Patrick Shipman, Sofya Chepushtanova, Eric Hanson, Francis Motta, and Lori Ziegelmeier. Persistence images: A stable vector representation of persistent homology. *Journal of Machine Learning Research*, 18(8):1–35, 2017.
- [2] Hirokazu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroya Inakoshi, Raphaël Tinarrage, and Yuhei Umeda. DTM-based filtrations. In *Topological Data Analysis (Abel Symposia)*, pages 33–66. Springer, 2020.
- [3] Rubén Ballester, Carles Casacuberta, and Sergio Escalera. Topological data analysis for neural network analysis: A comprehensive survey. *arXiv preprint arXiv:2312.05840*, 2023.
- [4] Peter Bubenik. Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, 16(1):77–102, 2015.
- [5] Mickaël Buchet, Frédéric Chazal, Steve Y Oudot, and Donald R Sheehy. Efficient and robust persistent homology for measures. *Computational Geometry*, 58:70–96, 2016.
- [6] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009.
- [7] Frédéric Chazal, David Cohen-Steiner, Leonidas J Guibas, Facundo Mémoli, and Steve Y Oudot. Gromov-Hausdorff stable signatures for shapes using persistence. *Computer Graphics Forum*, 28(5):1393–1403, 2009.
- [8] Frédéric Chazal, David Cohen-Steiner, and Quentin Mérigot. Geometric inference for probability measures. *Foundations of Computational Mathematics*, 11(6):733–751, 2011.
- [9] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

- [10] Vin de Silva and Gunnar Carlsson. Topological estimation using witness complexes. In *Symposium on Point-Based Graphics*, 2004.
- [11] Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1):12140, 2017.
- [12] Brittany Terese Fasy, Fabrizio Lecci, Alessandro Rinaldo, Larry Wasserman, Sivaraman Balakrishnan, and Aarti Singh. Confidence sets for persistence diagrams. *The Annals of Statistics*, 42(6):2301–2339, 2014.
- [13] Thomas Gebhart and Paul Schrater. Adversary detection in neural networks via persistent homology. *arXiv preprint arXiv:1711.10056*, 2017.
- [14] Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.
- [15] Feng Liu, Wenkai Xu, Jie Lu, Guangquan Zhang, Arthur Gretton, and Danica J Sutherland. Learning deep kernels for non-parametric two-sample tests. In *International Conference on Machine Learning (ICML)*, 2020.
- [16] Xingjun Ma, Bo Li, Yisen Wang, Sarah M Erfani, Sudanthi Wijewickrema, Grant Schoenebeck, Dawn Song, Michael E Houle, and James Bailey. Characterizing adversarial subspaces using local intrinsic dimensionality. In *International Conference on Learning Representations (ICLR)*, 2018.
- [17] Kanti V Mardia. Measures of multivariate skewness and kurtosis with applications. *Biometrika*, 57(3):519–530, 1970.
- [18] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. Topology of deep neural networks. *Journal of Machine Learning Research*, 21(184):1–40, 2020.
- [19] Stephan Rabanser, Stephan Günnemann, and Zachary C Lipton. Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1394–1406, 2019.
- [20] Donald R Sheehy. Linear-size approximations to the Vietoris–Rips filtration. *Discrete & Computational Geometry*, 49(4):778–796, 2013.
- [21] Gábor J Székely and Maria L Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.